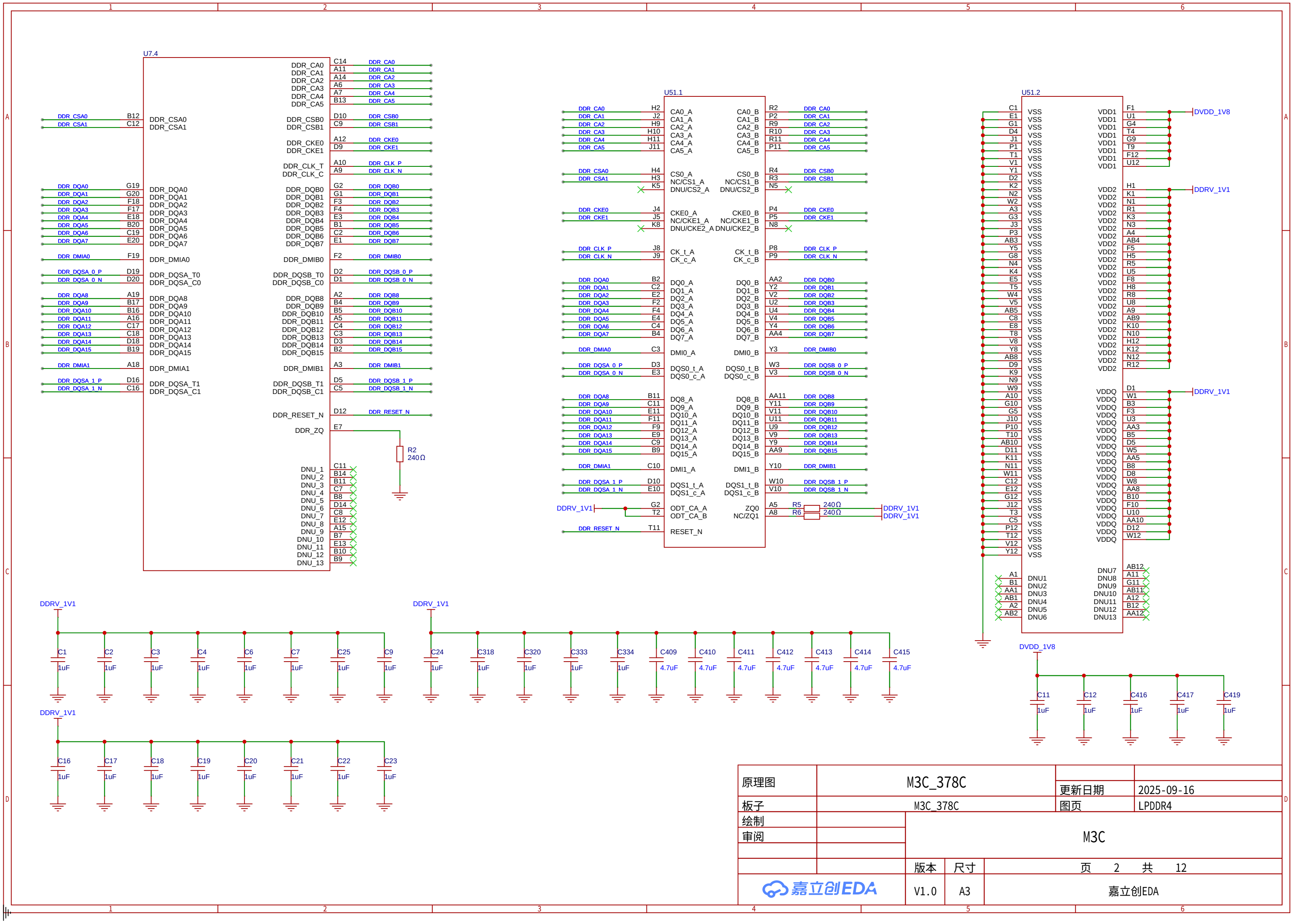


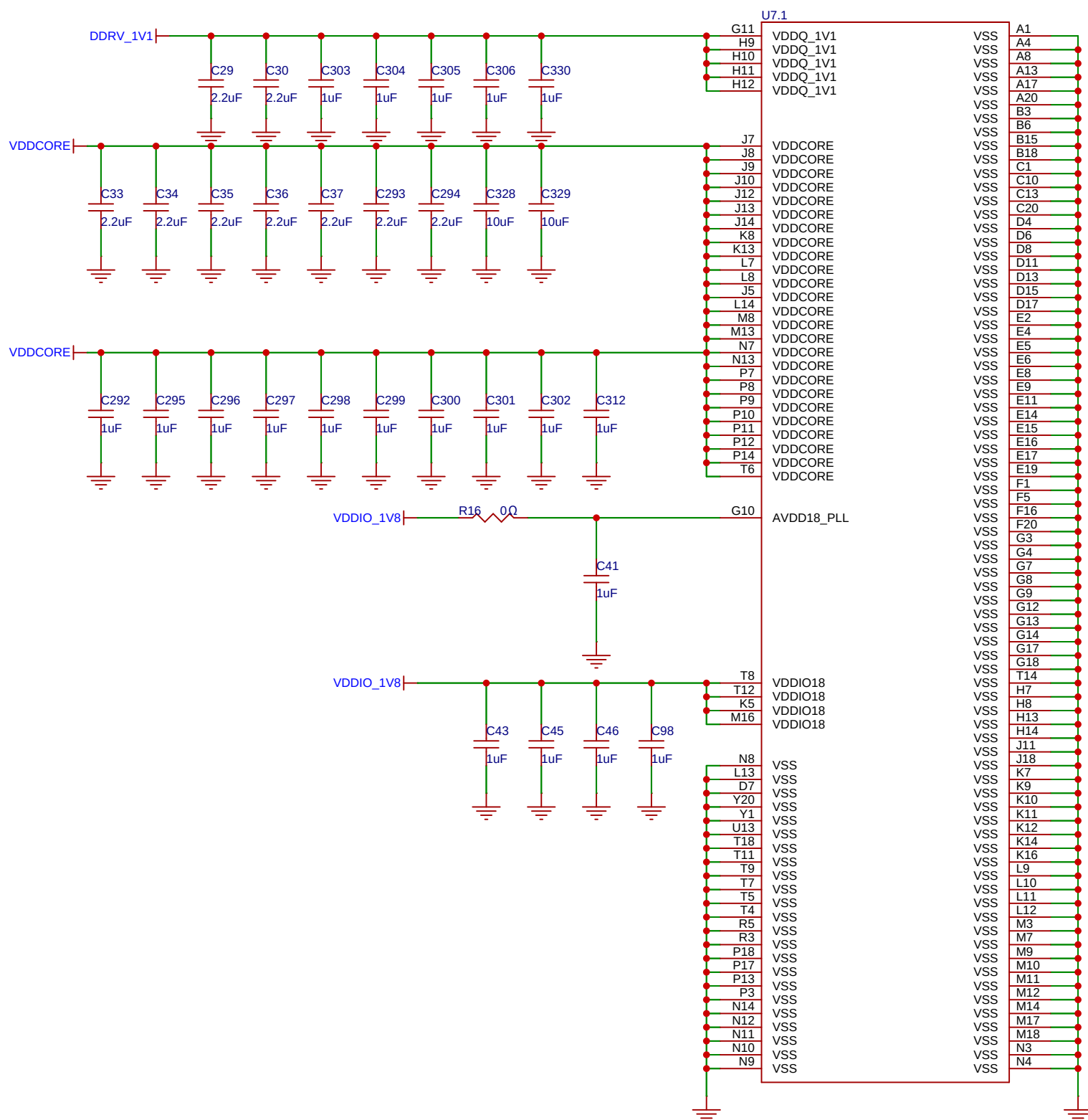
Revision History

Version	Date	Author	Description
378C	2025-09-16	SIPEED	Create Project

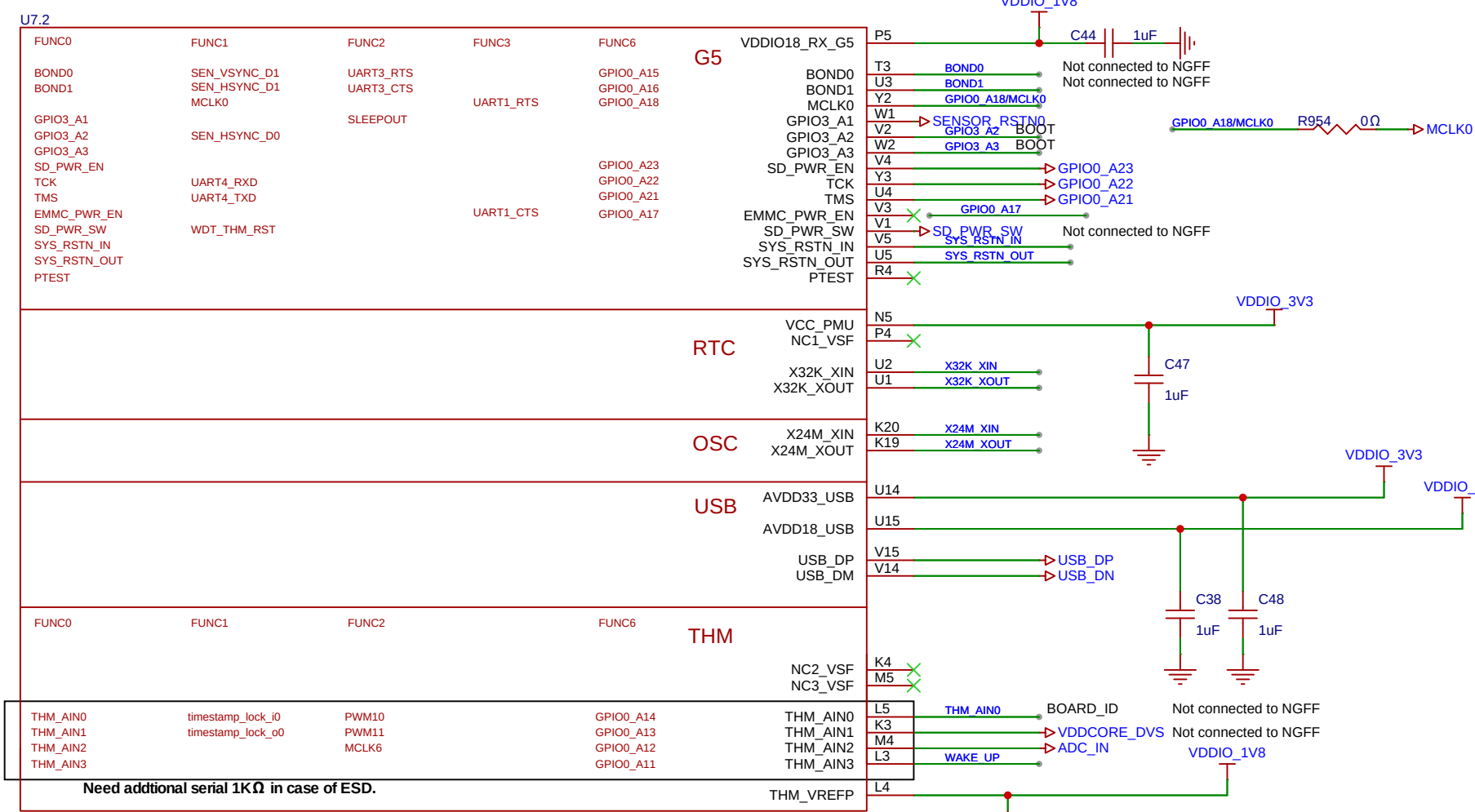
原理图	M3C_378C				
				更新日期	2025-09-16
板子	M3C 378C			图页	01 Revision History
绘制		M3C			
审阅					
		版本	尺寸	页 1 共 12	
		V1.0	A4	嘉立创EDA	



原理图	M3C_378C		更新日期	2025-09-16
板子	M3C 378C		图页	LPDDR4
绘制	M3C			
审阅				
	版本	尺寸	页 2 共 12	
嘉立创EDA		V1.0	A3	嘉立创EDA



原理图	M3C_378C			更新日期	2025-09-16
板子	M3C_378C			图页	Decap
绘制		M3C			
审阅					
		版本	尺寸	页 3 共 12	
		V1.0	A3	嘉立创EDA	

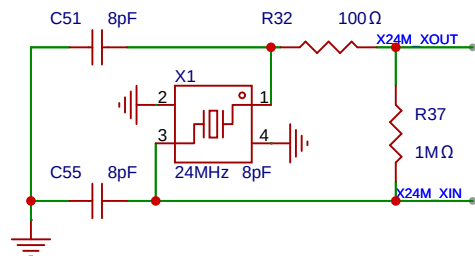
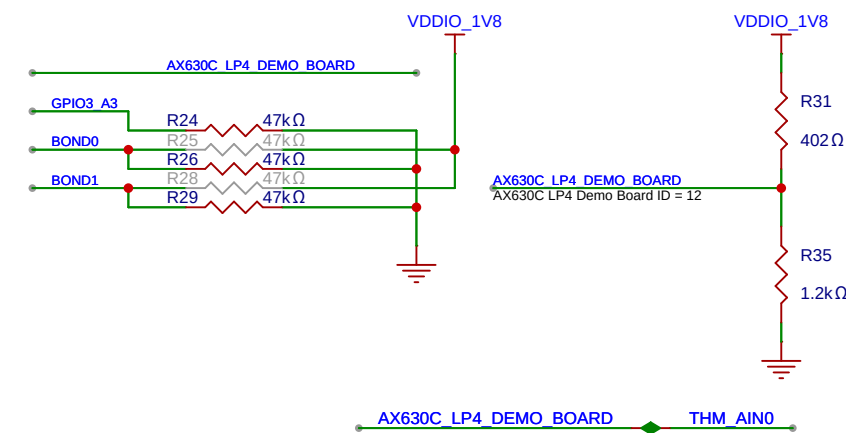


BOND1	BOND0	GPIO3_A3	GPIO3_A2	BOOT OPTION
0	1	1	1	NAND 2K
1	0	1	1	NAND 4K
1	1	1	1	NOR
0	0	0	1	EMMC UDA
0	1	0	1	SPI SLAVE
X	X	X	0	USB DL or SD Card or UART
X	X	X	X	UART DL

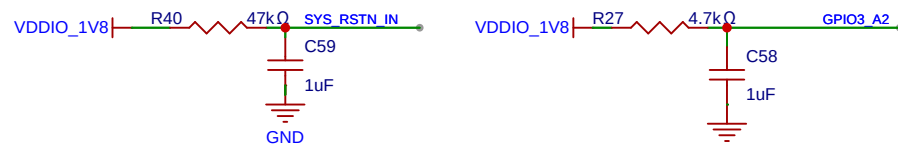
备注：如果需要NOR/NAND启动，需要在底板上面放GPIO3_A3上拉电阻，以及把R24改为NC

Note: If NOR/NAND boot is required, a pull-up resistor must be placed on GPIO3_A3 on the baseboard, and R24 should be changed to NC.

BOOT CONFIG

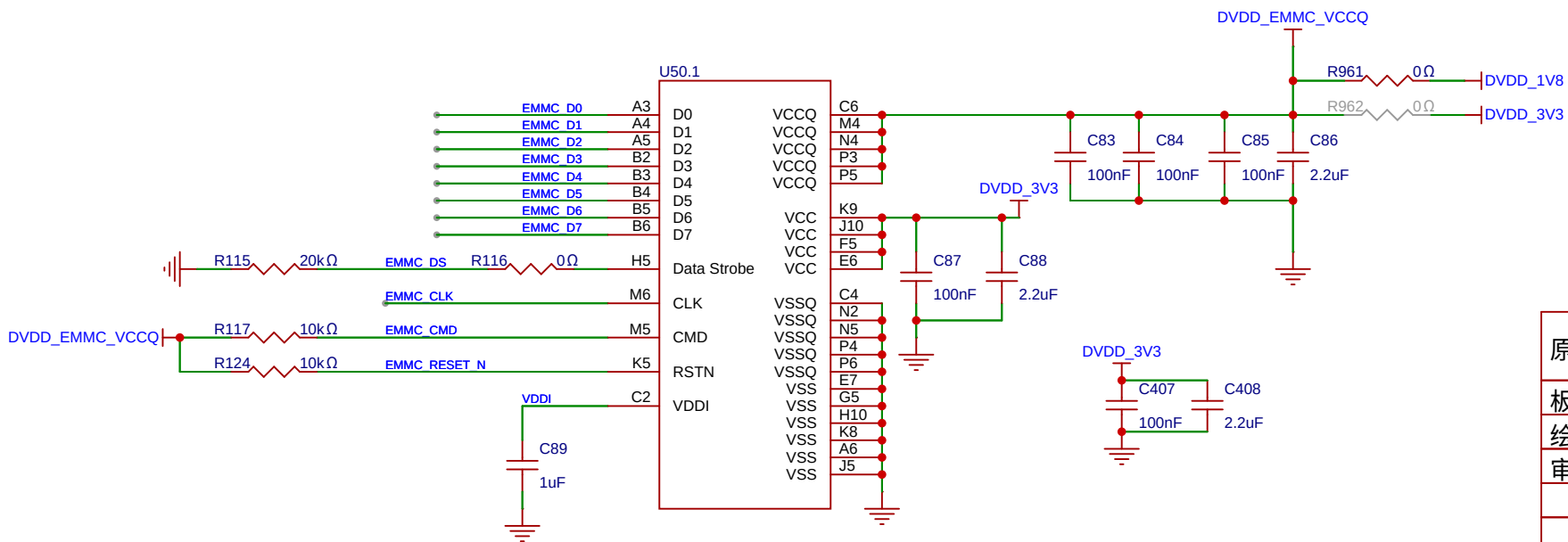
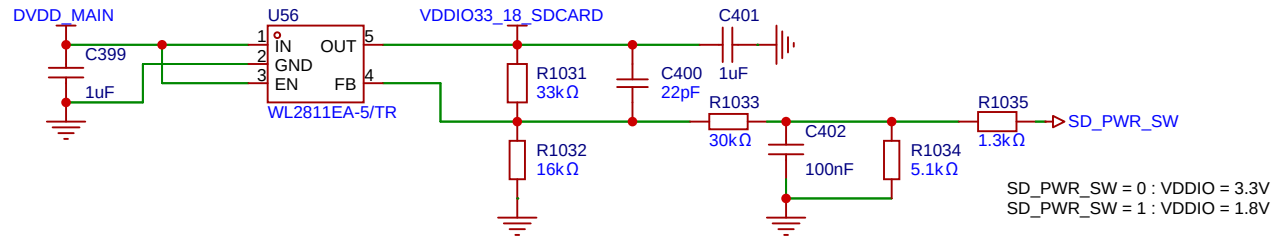
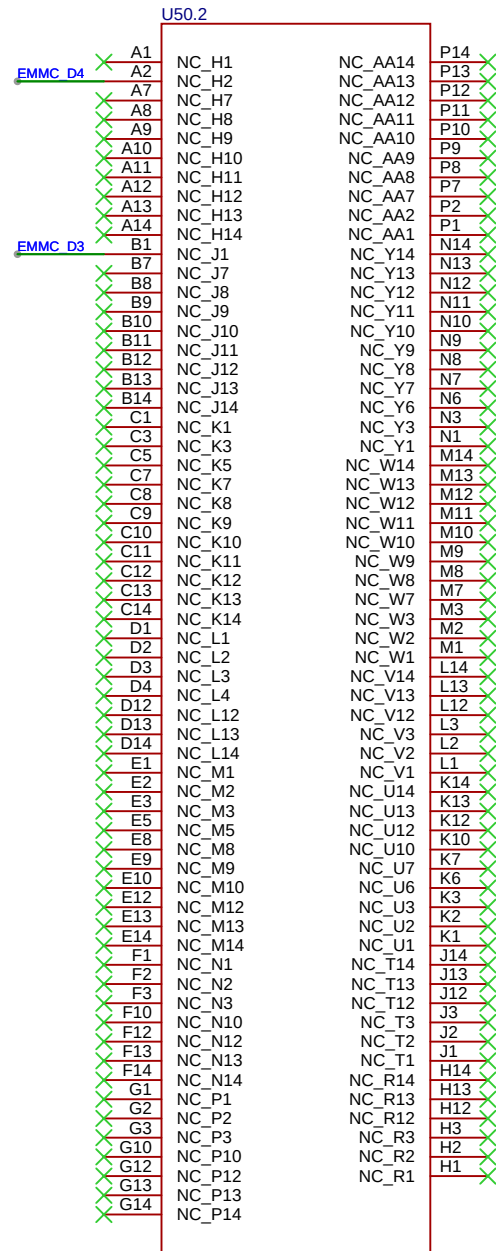
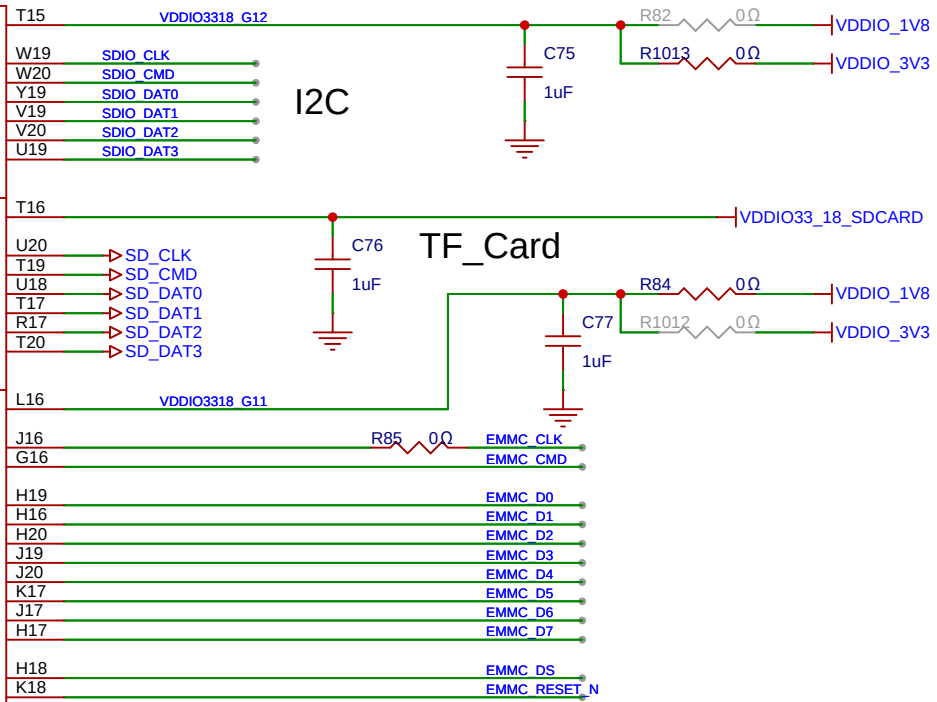
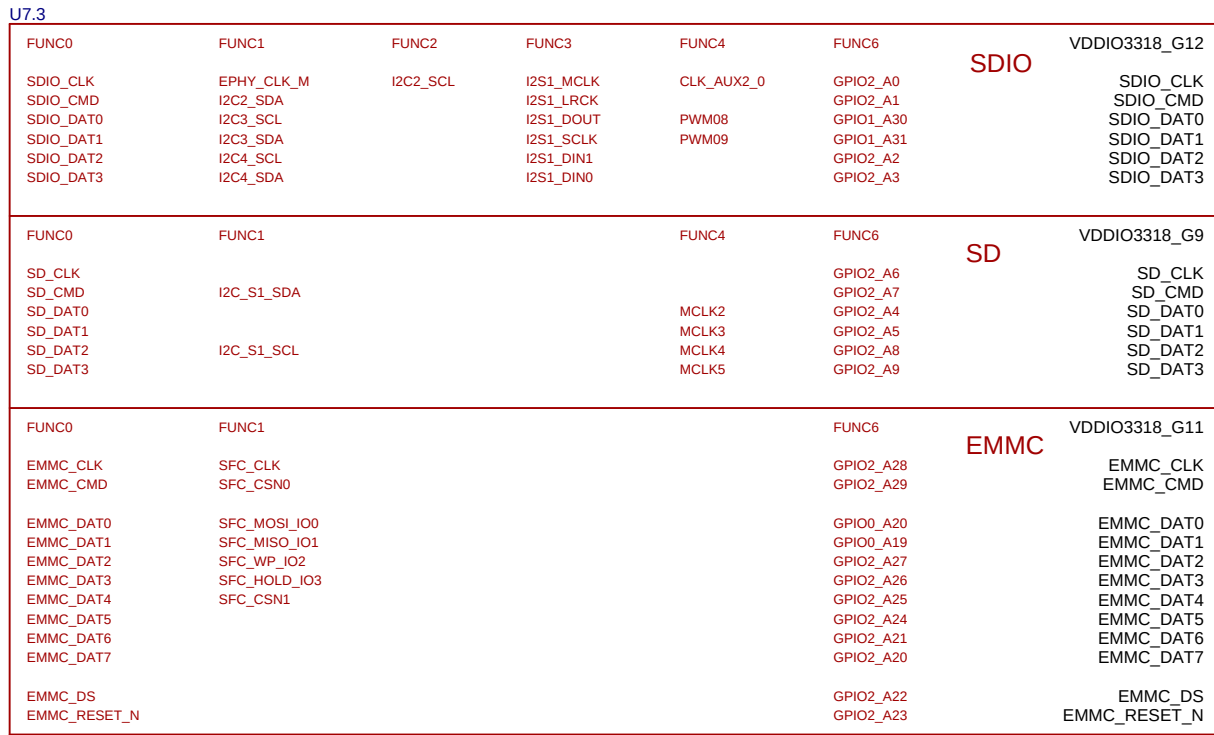


Download Button



Board ID	VREFP=1.8V	THM_AIN	
	ADC DATA[9:0]	R_Up	R_Down
0	0x000~0x020	NC	0
1	0x020~0x060	1.5K	100
2	0x060~0x0A0	1.4K	200
3	0x0A0~0x0E0	1.3K	300
4	0x0E0~0x120	1.2K	402
5	0x120~0x160	1.1K	510
6	0x160~0x1A0	1K	604
7	0x1A0~0x1E0	909	715
8	0x1E0~0x220	806	806
9	0x220~0x260	715	909
10	0x260~0x2A0	604	1K
11	0x2A0~0x2E0	510	1.1K
12	0x2E0~0x320	402	1.2K
13	0x320~0x360	300	1.3K
14	0x360~0x3A0	200	1.4K
15	0x3A0~0x3E0	100	1.5K
16	0x3E0~0x3FF	0	NC

原理图	M3C_378C		更新日期	2025-09-16
板子	M3C 378C		图页	SYS
绘制		M3C		
审阅				
		版本	尺寸	页 4 共 12
嘉立创EDA		V1.0	A3	嘉立创EDA



原理图	M3C_378C			更新日期	2025-09-16
板子	M3C_378C			图页	Flash
绘制		M3C			
审阅					
		版本	尺寸	页 5 共 12	
嘉立创EDA		V1.0	A3	嘉立创EDA	

U7.5

FUNC0	FUNC1	FUNC2	FUNC3	FUNC4	FUNC6
RGMII_MDCK	DPI_D1	PWM5_M	SEN_FLASH_D1_M		GPIO1_A24
RGMII_MDIO	DPI_D2	PWM6_M	SEN_FLASH_D2_M		GPIO1_A25
EPHY_CLK	DPI_PCLK	CLK_AUX1_2	SEN_FLASH_D3_M		GPIO1_A26
EPHY_RSTN	DPI_D0	PWM7_M	SEN_FLASH_D4_M	SPI_m2_CS0_M	GPIO1_A27
RGMII_RXCLK	DPI_D3				GPIO1_A15
RGMII_RXDV	DPI_D4				GPIO1_A14
RGMII_RXD0	DPI_D5			UART0_CTS	GPIO1_A12
RGMII_RXD1	DPI_D6			UART0_RTS	GPIO1_A13
RGMII_RXD2	DPI_D7				GPIO1_A16
RGMII_RXD3	DPI_D8				GPIO1_A17
RGMII_TXCLK	DPI_D9		SEN_HSYNC_D1_M	SPI_m2_SCLK_M	GPIO1_A20
RGMII_TXEN	DPI_D10		SEN_VSYNC_D0_M	SPI_m2_CS1_M	GPIO1_A21
RGMII_TXD0	DPI_D11		MCLK1	SPI_m2_MOSI_M	GPIO1_A18
RGMII_TXD1	DPI_D12		SEN_HSYNC_D0_M	SPI_m2_MISO_M	GPIO1_A19
RGMII_TXD2	DPI_D13	PWM3_M	SEN_VSYNC_D1_M		GPIO1_A22
RGMII_TXD3	DPI_D14	PWM4_M	SEN_FLASH_D0_M		GPIO1_A23
EMAC_PTP_PPS0	DPI_D17	PWM0_M	UART5_CTS	SEN_FLASH_D5	GPIO1_A8
EMAC_PTP_PPS1	DPI_D16	PWM1_M	UART5_RTS	SEN_VSYNC_D0	GPIO1_A9
EMAC_PTP_PPS2	DPI_D18		UART5_TXD		GPIO1_A10
EMAC_PTP_PPS3	DPI_D15	PWM2_M	UART5_RXD		GPIO1_A11
EPHY_LED0	RGMII_MDCK_M	TE0	SEN_FLASH_D5_M	SPI_m2_CS3_M	GPIO1_A28
EPHY_LED1	RGMII_MDIO_M	TE1	SEN_ELEC_PLS_M	SPI_m2_CS2_M	GPIO1_A29

G8

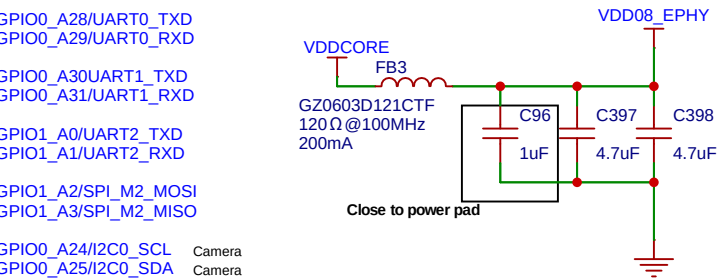
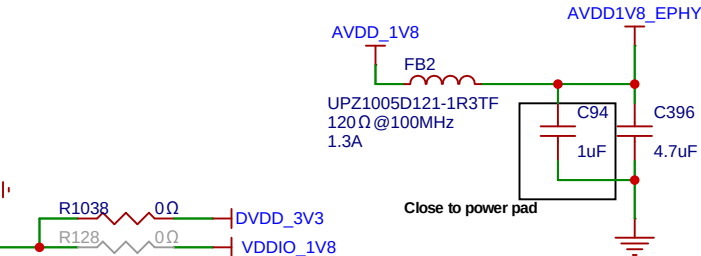
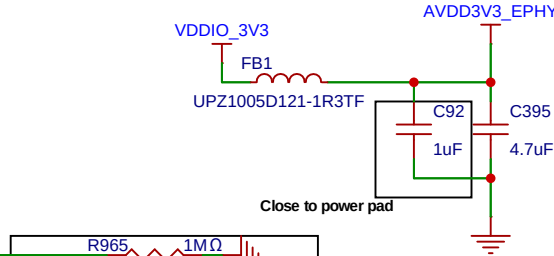
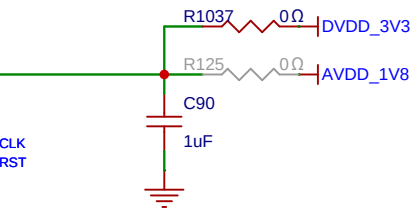
VDDIO3318_G8
RGMII_MDCK
RGMII_MDIO
EPHY_CLK
EPHY_RSTN
RGMII_RXCLK
RGMII_RXDV
RGMII_RXD0
RGMII_RXD1
RGMII_RXD2
RGMII_RXD3
RGMII_TXCLK
RGMII_TXEN
RGMII_TXD0
RGMII_TXD1
RGMII_TXD2
RGMII_TXD3
EMAC_PTP_PPS0
EMAC_PTP_PPS1
EMAC_PTP_PPS2
EMAC_PTP_PPS3
EPHY_LED0
EPHY_LED1

EPHY

AVDD33_EPHY
AVDD18_EPHY
AVDD08_EPHY
EPHY_RXP
EPHY_RXN
EPHY_TXP
EPHY_TXN
EPHY_RTX

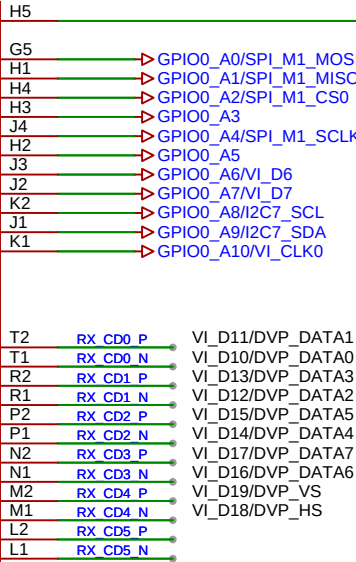
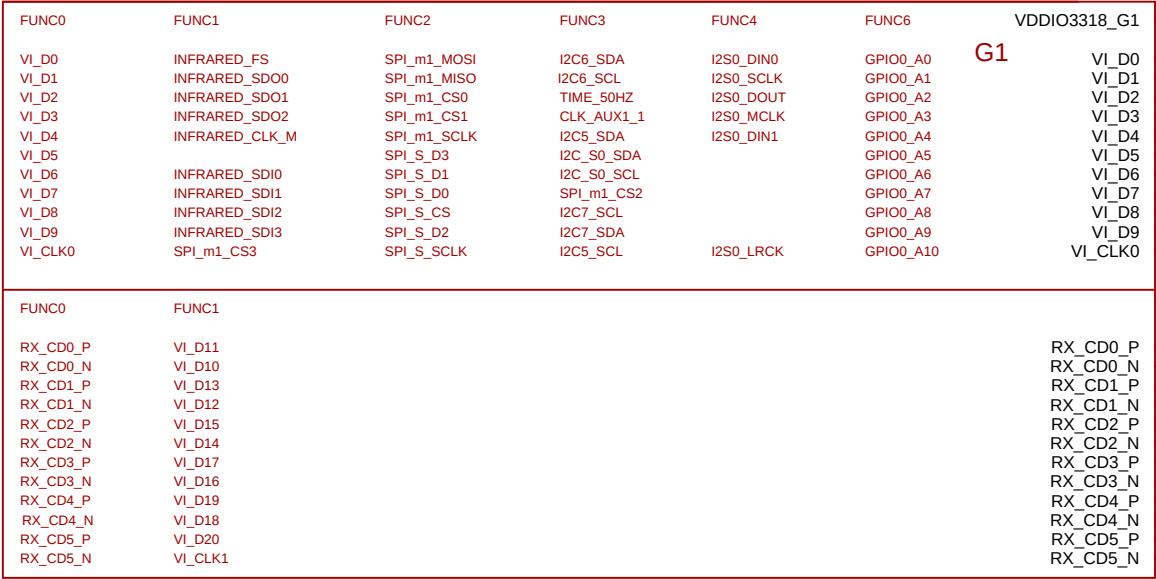
G6

VDDIO3318_G6
UART0_TXD
UART0_RXD
UART1_TXD
UART1_RXD
UART2_TXD
UART2_RXD
UART3_TXD
UART3_RXD
I2C0_SCL
I2C0_SDA
I2C1_SCL
I2C1_SDA



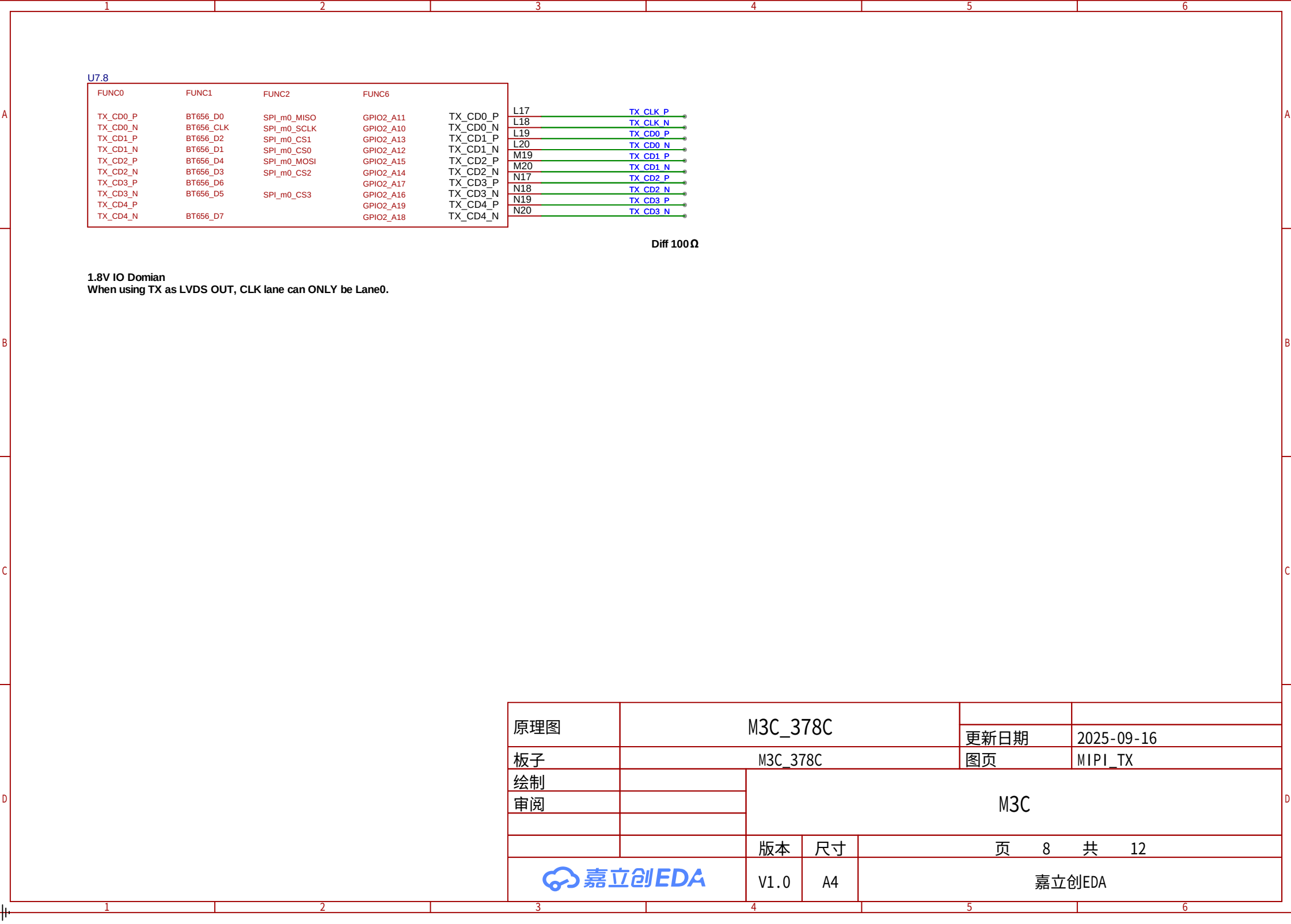
原理图	M3C_378C		更新日期	2025-09-16
板子	M3C 378C		图页	ETH_I0
绘制		M3C		
审阅				
		版本	尺寸	页 6 共 12
嘉立创EDA		V1.0	A3	嘉立创EDA

U7.7

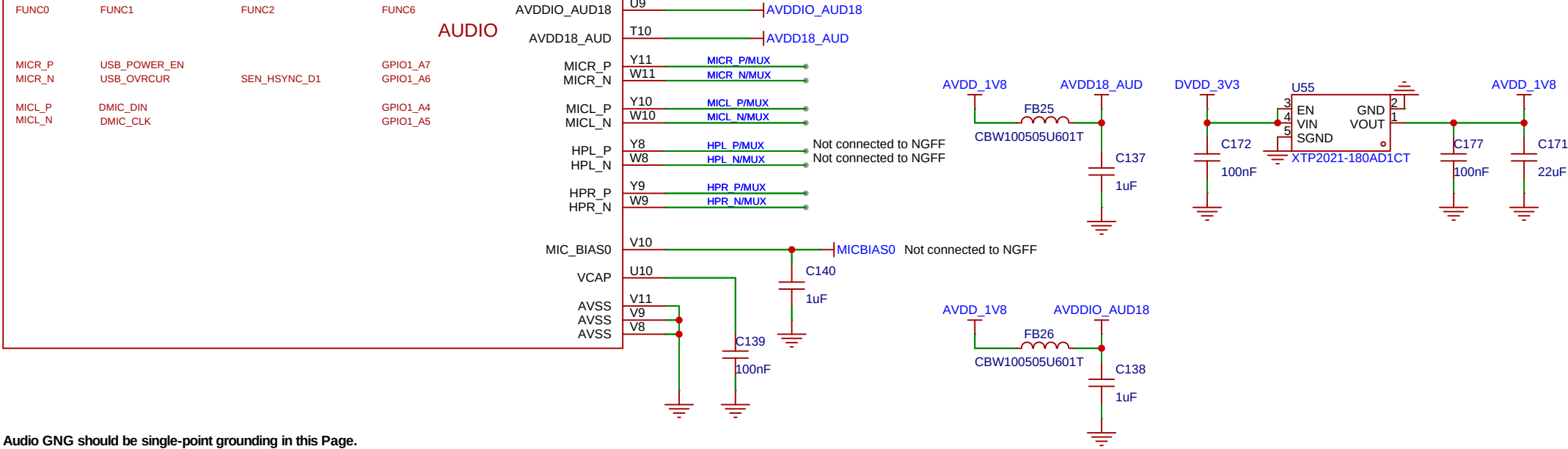


1.8V RX IO Domian
Supplied by VDDIO18_RX_G5 (PIN P5)

原理图	M3C_378C			更新日期	2025-09-16
板子	M3C_378C			图页	MIPI_RX
绘制		M3C			
审阅					
		版本	尺寸	页 7 共 12	
嘉立创EDA		V1.0	A4	嘉立创EDA	

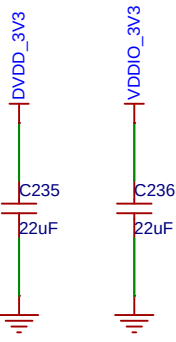
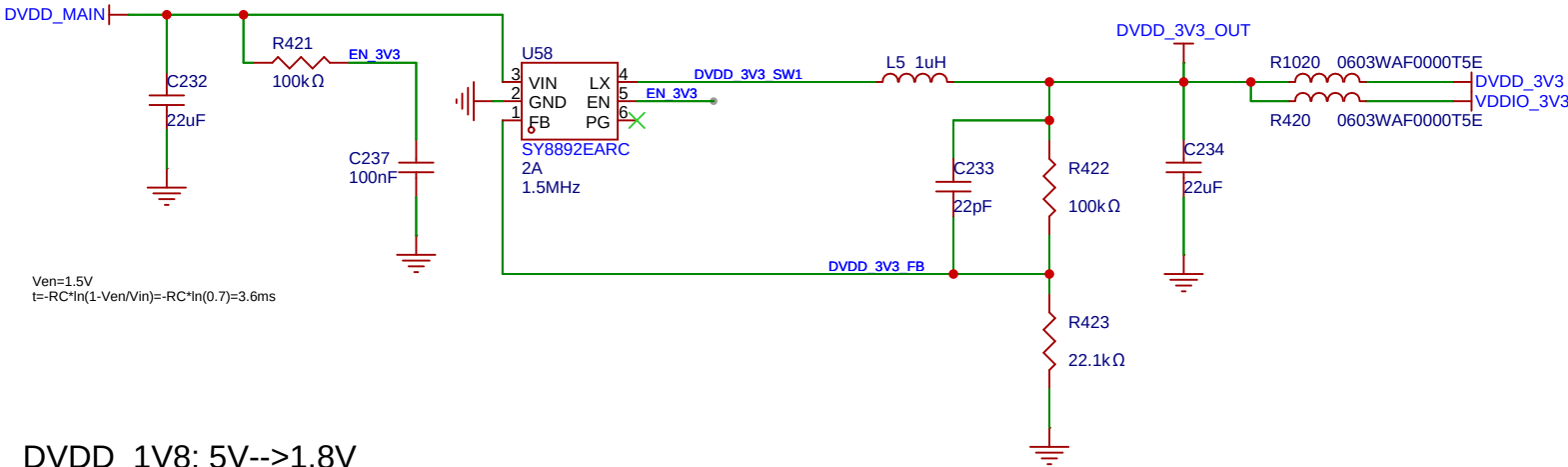


U7.6

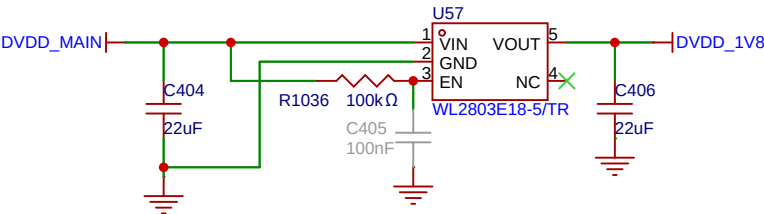
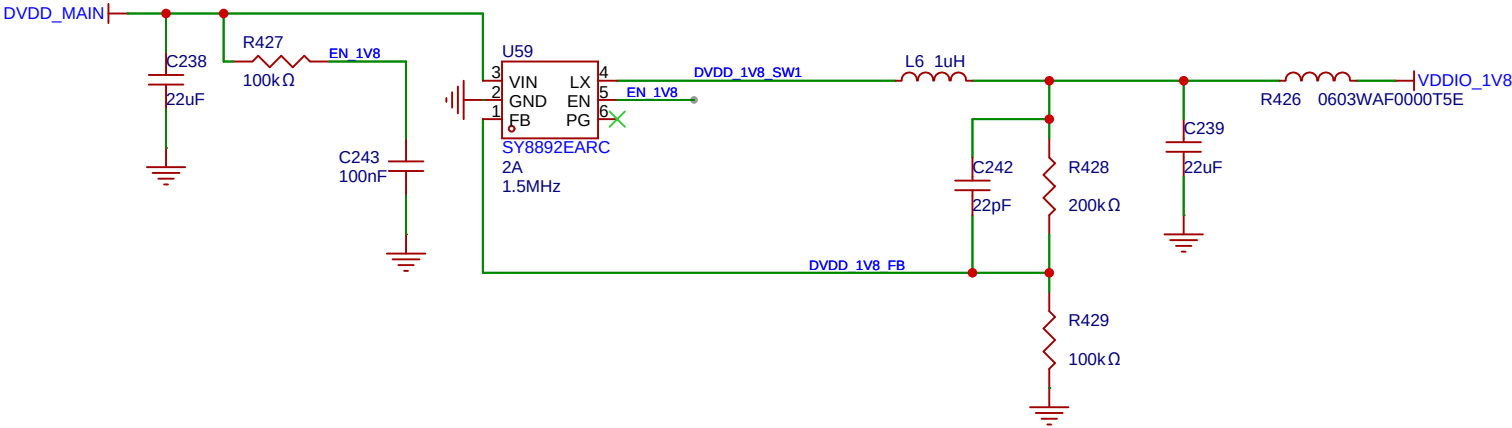


原理图	M3C_378C				
				更新日期	2025-09-16
板子	M3C_378C			图页	Codec
绘制		M3C			
审阅					
		版本	尺寸	页 9 共 12	
		V1.0	A4	嘉立创EDA	

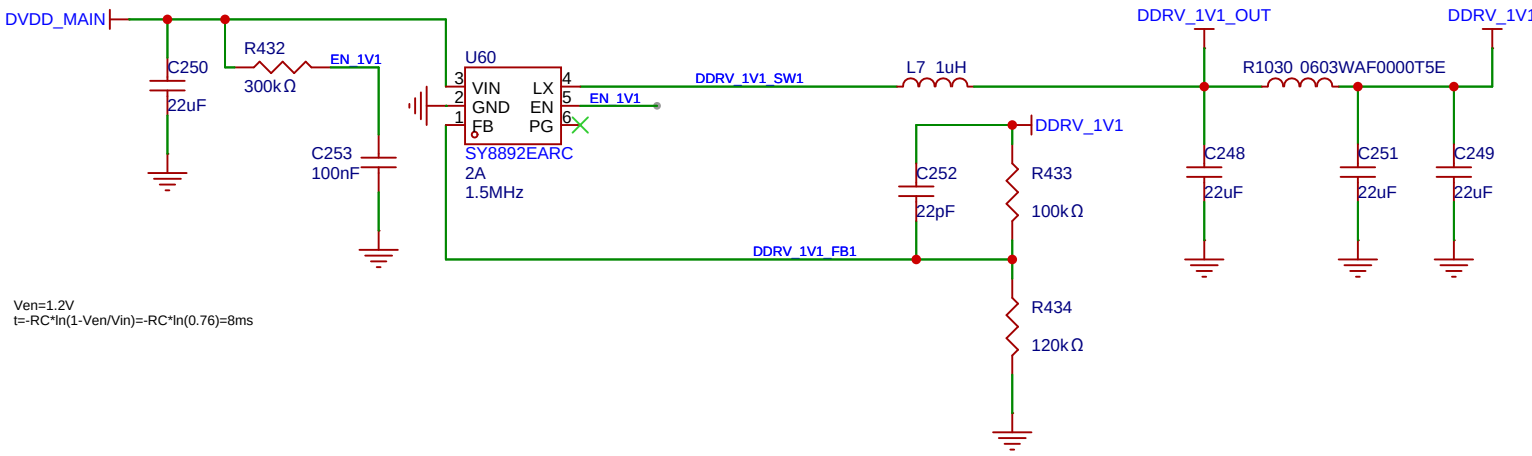
DVDD_3V3: 5V-->3.3V
 $V_{out}=0.6 \times (1+100K/22.1K)=3.3V$



DVDD_1V8: 5V-->1.8V
 $V_{out}=0.6 \times (1+200K/100K)=1.8V$

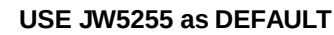
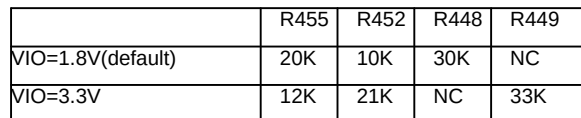


DDRV_1V1: 5V-->1.1V
 $V_{out}=0.6 \times (1+100K/120K)=1.1V$



$V_{en}=1.2V$
 $t=-RC \ln(1-V_{en}/V_{in})=-RC \ln(0.76)=8ms$

原理图	M3C_378C		更新日期	2025-09-16
板子	M3C_378C		图页	DCDC_IO_DDR
绘制		M3C		
审阅				
		版本	尺寸	页 10 共 12
嘉立创EDA		V1.0	A3	嘉立创EDA

$$V_{out}=0.6*(1+47K/120K)=0.835V$$

$$\text{VDDCORE_DVS Duty Cycle} = (5.4 - 6 * \text{VDDCORE}) / 1.2 * 100\%$$


原理图	M3C_378C			更新日期	2025-09-16
板子	M3C_378C			图页	DCDC CORE
绘制		M3C			
审阅					
		版本	尺寸	页	11 共 12
		V1.0	A4	嘉立创EDA	



原理图	M3C_378C				
				更新日期	2025-09-16
板子	M3C_378C			图页	PinOut
绘制		M3C			
审阅					
		版本	尺寸	页	12 共 12
		V1.0	A4	嘉立创EDA	